AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

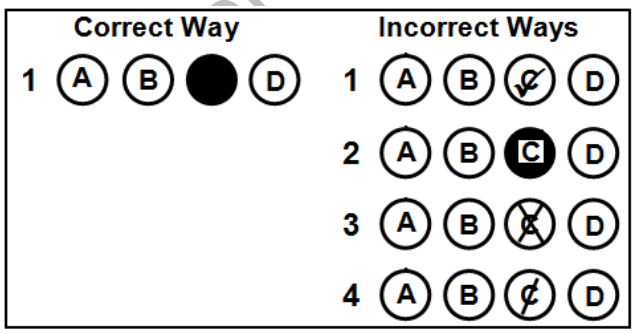
Chemistry Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The decomposition products of $PbCl_4$ at room temperature are

   A. $Pb + Cl_4$
   B. $Pb + 2Cl_2$
   C. $PbCl_2 + 2Cl^-$
   D. $PbCl_2 + Cl_2$

2. The weakest hydrohalic acid is

   A. hydroiodic acid.
   B. hydrofluoric acid.
   C. hydrochloric acid.
   D. hydrobromic acid.

3. The oxide of period 3 that will produce an acidic solution in an aqueous medium is

   A. Al_2O_3
   B. MgO
   C. P_4O_{10}
   D. Na_2O

4. The first ionisation energy generally decreases down the group in the periodic table because there is an increase in

   I. atomic radius
   II. nuclear charge
   III. electron affinity
   IV. number of screening electrons

   A. I and II.
   B. I and IV.
   C. II and III.
   D. III and IV.

5. All of the following trends are observed in period 3 elements when moving from left to right in the periodic table EXCEPT a gradual

   A. increase in their melting point.
   B. change in their oxides from basic to acidic.
   C. change in the type of bonding from ionic to covalent.
   D. decrease in atomic radii with constant shielding effect.

6. If the coordination sphere carries a negative charge, then the suffix for transition metal ion in International Union of Pure and Applied Chemistry (IUPAC) nomenclature would be

   A. ate.
   B. ane.
   C. ene.
   D. ium.

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7. Cu^{2+} ion is more stable than Cu^{+} ion in an aqueous solution because Cu^{2+} ion has

(**Note**: $^{63.5}_{29}Cu$)

A. half-filled d-orbital.
B. greater charge density.
C. lower hydration energy.
D. completely filled d-orbital.

8. In forming coordination complexes, all of the following species would act as a ligand EXCEPT

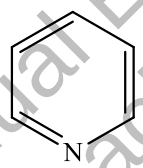
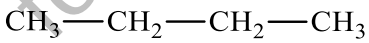
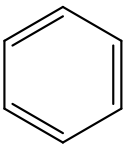
A. Cl^-
B. CO
C. OH^-
D. NH_4^+

9. The electronic structure of iron(II) ion is

(**Note**: $^{56}_{26}Fe$)

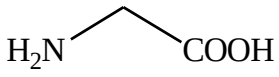
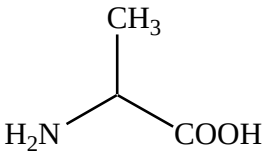
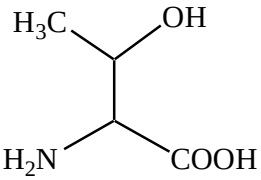
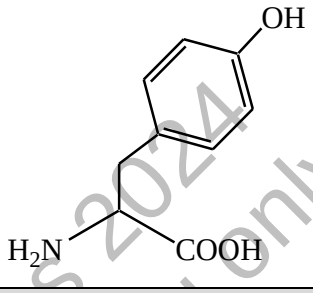
A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$
B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$
C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$
D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^4$

10. The compound which is alicyclic in nature is

	
A	B
	
C	D

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11. All of the following amino acid molecules exhibit stereoisomerism EXCEPT

	
A	B
	
C	D

12. The compound that contains a carbonyl group is

A. $C_2H_5OC_2H_5$
B. CH_3CH_2OH
C. $C_2H_5CO_2CH_3$
D. $CH_3C(OH)_2CH_3$

13. In the nitration of benzene, the reaction between nitric acid and sulphuric acid generates an electrophile that is

A. nitrite.
B. nitrate.
C. nitroso.
D. nitronium.

14. The reaction of Cl_2 with toluene in dark, in the presence of a catalyst ($FeCl_3$) and at low temperatures yields

A. p-chlorotoluene only.
B. m-chlorotoluene only.
C. a mixture of o-chlorotoluene and p-chlorotoluene.
D. a mixture of p-chlorotoluene and m-chlorotoluene.

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15. Consider the following structural features of a class of compounds.

- Each carbon is sp^3 hybridised.
- Complete rotation is restricted around the single bond.
- Constitutional isomers are possible.
- The most stable form contains six carbon atoms.

The class of compounds that exhibits these features is

A. alkanes.
B. alkenes.
C. cycloalkanes.
D. substituted benzene.

16. In a molecule of ethyne, the total numbers of sigma (σ) and pi (π) bonds are

	Number of	
	Sigma (σ) Bond	**Pi (π) Bond**
A	1	4
B	4	1
C	3	2
D	2	3

17. The compound which can be used as an alternate to chlorofluorocarbons in refrigeration is

A. methyl chloride.
B. difluoromethane.
C. carbon tetrachloride.
D. trichlorofluoromethane.

18. Smog is a combination of smoke and fog.

Smog is chemically reducing in nature if it contains a high concentration of

A. ozone.
B. nitric oxide.
C. sulphur dioxide.
D. unburnt hydrocarbons.

19. One of the parameters to check water purity is electrical conductivity which measures the amount of

A. ions in water.
B. dissolved oxygen in water.
C. molecular motion in water.
D. suspended particles in water.

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20. The application of green chemistry in the industrial sector does NOT help to

A. reduce hazardous waste.
B. reduce the use of petroleum products.
C. increase consumerism by making safer products.
D. increase the yield of non-biodegradable products.

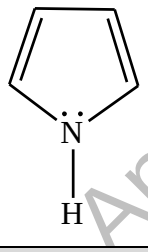
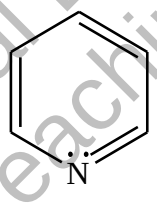
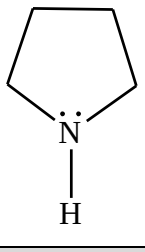
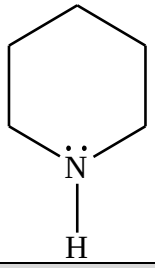
21. All of the following are true about S_N2 reactions EXCEPT that they

A. are favoured in non-polar solvents.
B. always occur with partial racemisation.
C. involve the formation of transition state.
D. always occur with inversion of configuration.

22. The polar nature of the N–H bond in amines results in

	Melting Point	**Boiling Point**
A	low	low
B	high	low
C	low	high
D	high	high

23. Pyridine is a basic, heterocyclic amine. Its structure is represented as

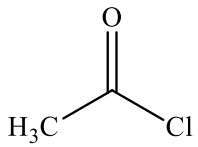
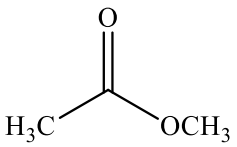
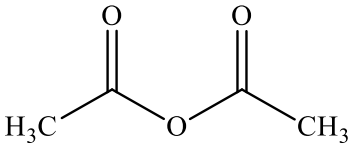
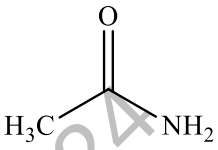
A	B	C	D
			

24. When butyl magnesium bromide reacts with carbon dioxide gas, it forms

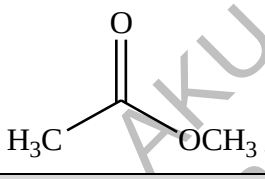
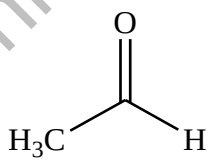
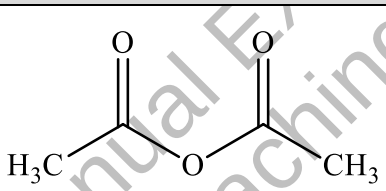
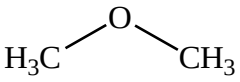
A. pentanal.
B. pentanol.
C. pentanone.
D. pentanoic acid.

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25. The carboxylic acid derivative which is the LEAST reactive is

	
A	B
	
C	D

26. The structure of organic compound which is prepared by replacing the active hydrogen of CH_3COOH with an alkyl group is

	
A	B
	
C	D

27. The given chemical reaction shows oxidation of methanol in two steps.

$$CH_3OH + [O] \xrightarrow{Na_2Cr_2O_7/H_2SO_4} \mathbf{X} + [O] \xrightarrow{Na_2Cr_2O_7/H_2SO_4} \mathbf{Y}$$

The resultant **X** and **Y** in these steps are

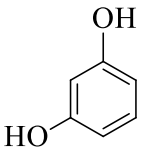
	X	**Y**
A	formaldehyde	formic acid
B	acetaldehyde	acetic acid
C	formic acid	formaldehyde
D	acetic acid	acetaldehyde

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28. The molecules of formic acid exist as dimers in vapour phase due to

A. the presence of intermolecular forces.
B. the process of polymerisation.
C. condensation reaction.
D. low boiling point.

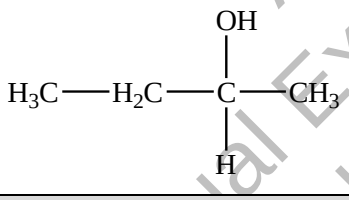
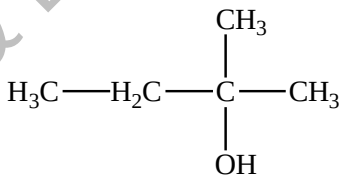
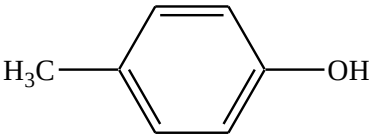
29. Given is the structure of resorcinol.



The International Union of Pure and Applied Chemistry (IUPAC) name of this compound is

A. o-Dihydroxyphenol.
B. p-Dihydroxyphenol.
C. 1,3-Dihydroxybenzene.
D. 1,5-Dihydroxybenzene.

30. Which of these compounds reacts immediately with Lucas reagent and forms an oily layer by turning the solution turbid?

	
A	B
	
C	D

31. Diazonium salt is formed as an intermediate during the preparation of phenol from

A. aniline.
B. cumene.
C. chlorobenzene.
D. benzene sulphonic acid.

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32. Which of the following sets of substances produces an ether through Williamson synthesis reaction?

A. $Na + CH_3OH$
B. $Na + CH_3O^-Na^+$
C. $C_2H_5Br + CH_3OH$
D. $C_2H_5Br + CH_3O^-Na^+$

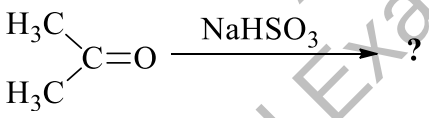
33. In cold state, $H_3C - H_2C - O - CH_2 - CH_3$ is unable to react with sodium metal because of the absence of

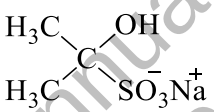
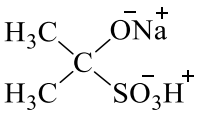
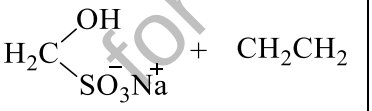
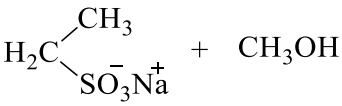
A. a hydroxyl group.
B. an acidic hydrogen.
C. an ionic functional group.
D. sp^3 hybridised oxygen atom.

34. Which of the following ketones will have the LOWEST boiling point?

A. CH_3COCH_3
B. $CH_3COCH_2CH_3$
C. $CH_3COCH_2CH_2CH_3$
D. $CH_3COCH_2CH_2CH_2CH_3$

35. The product(s) of the given reaction is/ are



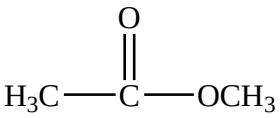
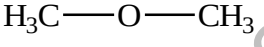
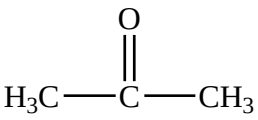
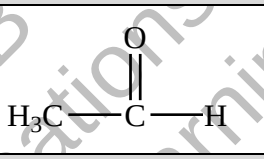
	
A	B
	
C	D

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36. An organic compound **X** undergoes two different reactions: oxidation and reduction and the results are as follows.

Reaction	Result
Oxidation	**X** is converted into CH_3COOH
Reduction	**X** is converted into CH_3CH_2OH

Based on these results, the structure of **X** is identified as

	
A	B
	
C	D

37. Consider the following characteristics.

I. It is non-reducing.
II. It is a disaccharide.
III. It gives a positive result with Fehling's test.
IV. It gives a positive result with Benedict's test.

The characteristics that represent fructose as a ketonic sugar are

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

38. A disproportionation reaction between two benzaldehyde molecules in the presence of sodium hydroxide base results in the formation of sodium benzoate and benzyl alcohol.

The chemical reaction described in the given statement is an example of

A. Cannizzaro's reaction.
B. Aldol condensation.
C. haloform reaction.
D. Grignard reaction.

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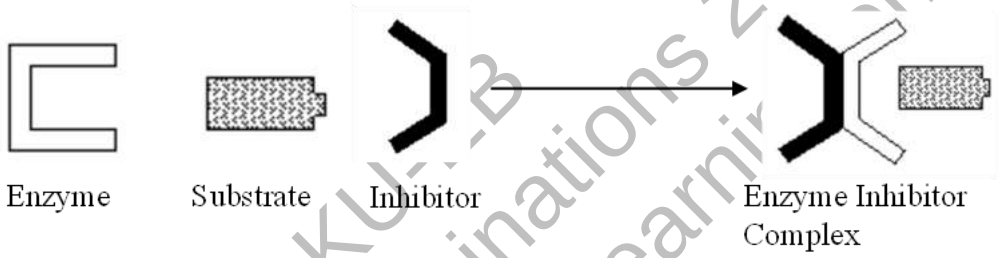
39. In contrast to ribonucleic acid (RNA), deoxyribonucleic acid (DNA) strands

A. are much shorter in length.
B. are unstable in alkaline conditions.
C. contain adenine and uracil as base pair.
D. contain pentose sugar with one less hydroxyl group.

40. The chemical which is released by the pancreatic glands after having a meal is

A. insulin.
B. estrogen.
C. androgen.
D. adrenaline.

41. The given diagram depicts a type of inhibition in an enzyme-catalysed reaction.



The inhibitor in this reaction is classified as non-competitive because it

A. resembles the shape of the enzyme.
B. attaches with the site other than active site.
C. breaks bond between the substrate and the enzyme.
D. modifies the shape of the enzyme to favour substrate binding.

42. The characteristic feature of the primary structure of a protein molecule is the

A. twisting of alpha and beta pleated sheets.
B. sequencing of amino acids in a polypeptide chain.
C. three-dimensional folding of the polypeptide chains.
D. bridging of two cysteine side chains via disulphide bonds.

43. All of the following are the significances of dyes EXCEPT that they are used in

A. laser scanning.
B. textile printing.
C. food processing.
D. biological staining.

44. Polyvinyl chloride (PVC) is used in the making of

A. cardigans.
B. explosives.
C. hose pipes.
D. boot polish.

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45. Consider the given characteristics of a class of organic compounds.

- Elastic in nature
- Derivatives of petrochemicals
- Made of monomers and polymers

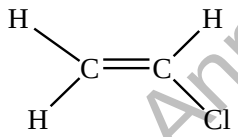
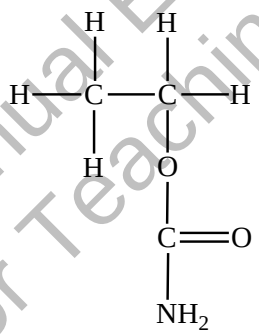
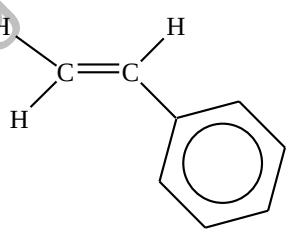
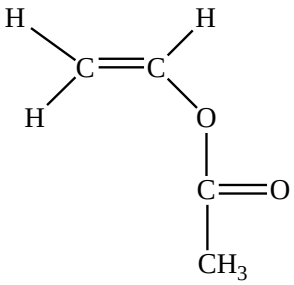
Based on these characteristics, the class of compounds is identified as

A. drugs.
B. fertilisers.
C. adhesives.
D. pesticides.

46. The pair that matches the component of paint with its appropriate purpose is

	Component	Purpose
A	volatile solvent	gives colour
B	pigment	provides viscosity
C	binder	creates a continuous film
D	clay	allows quick drying of paint

47. Which of the following monomers does NOT undergo free radical polymerisation to form thermoplastics?

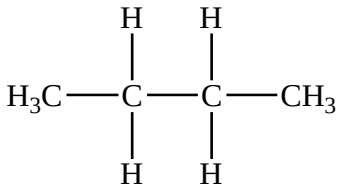
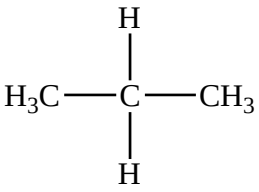
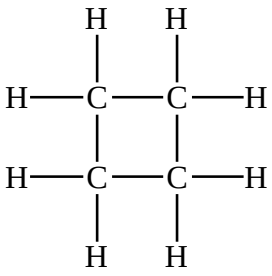
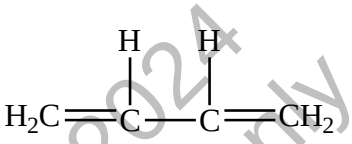
A	B	C	D
			

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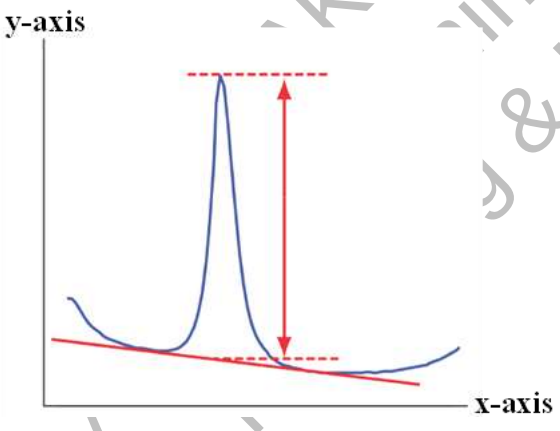
48. The CORRECT sequence of the five stages in the working of a mass spectrometer is

	Stages				
	I	II	III	IV	V
A	acceleration	deflection	detection	ionisation	vapourisation
B	ionisation	deflection	vapourisation	acceleration	detection
C	detection	acceleration	deflection	ionisation	vapourisation
D	vapourisation	ionisation	acceleration	deflection	detection

49. The molecule that will absorb in the ultraviolet-visible region of electromagnetic spectrum is

	
A	B
	
C	D

50. Consider the given graph of an atomic emission spectroscopy.



Based on the spectrum, the variables on x and y axes are

	x-axis	y-axis
A	concentration	emission intensity
B	wavelength	absorbance
C	concentration	absorbance
D	wavelength	emission intensity

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